



ASSIGNMENT 2

GROUP 3

SECV2113-01 HUMAN COMPUTER INTERACTION

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Table of contents

1.0 Introduction	3
1.1 Nielsen's heuristics consist of the following 10 rules:.....	3
1.2 Severity Rating	4
2.0 Method of evaluation	5
3.0 Heuristic Evaluation (HE) Table	6
4.0 User journey map.....	11

1.0 Introduction

In Assignment 2 of our Human-Computer Interaction course, we will conduct heuristic evaluations on the paper prototype designed in Project Part 4, which serves as a prototype for our feedback system. This evaluation marks a crucial step in the project's overall progression, enabling us to critically assess our design using established usability principles. We will apply **Nielsen's heuristics**, a set of 10 rules, to evaluate the user interfaces that have created. By assessing these interfaces and rating the severity level of any observed issues, we aim to identify areas for improvement and enhance the overall user experience.

1.1 Nielsen's heuristics consist of the following 10 rules:

1. Visibility of system status. – Ensuring that users receive information in real time, which the system should continuously inform users about what is going on through appropriate feedback within a reasonable time.

2. Match between system and real world. – Making the user interface recognisable and easy to use, which the system should speak the users' language, with words, phrases, and concepts familiar to the user, rather than system-oriented terms.

3. User control and freedom. – Multifunctional application that makes it simple for users to engage and navigate the system. For example, users often choose system functions by mistake and need a clearly marked “emergency exit” to leave the unwanted state without having to go through an extended dialogue.

4. Consistency and standards. – Ensuring consistency and satisfying user expectations while designing the interface. For instance, the users should not have to wonder whether different words, situations, or actions mean the same thing by following platform conventions.

5. Error prevention. – Trigger warning by giving alerts or suggestions for certain actions to avoid user errors. It means a good error message is a careful design that prevents a problem from occurring in the first place.

6. Recognition rather than recall. – Facilitating users' recognition and use of various system functionalities. For example, minimising the user's memory load by making objects, actions, and

options visible. The user should not have to remember information from one part of the dialogue to another.

7. Flexibility and efficiency of use. – Shortcuts and effective solutions are offered to improve user experience. For instance, accelerators may often speed up the interaction for the expert user such that the system can cater to both inexperienced and experienced users.

8. Aesthetic and minimalist design. – Buttons, elements and features that are relevant and visually appealing are designed. For example, the app should not contain information which is irrelevant or rarely needed. Every extra unit of information in a system competes with the relevant units of information and reduces their relative visibility.

9. Help users recognize, diagnose, and recover from errors. – Assisting user in understanding to identify and resolve the errors, which the error messages should be expressed in plain language without codes, precisely indicate the problem, and constructively suggest a solution.

10. Help and documentation. – Helpful instructions and information are provided to guide the users. For instance, any such information should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too large.

1.2 Severity Rating

1. Cosmetic Issue
2. Minor Issue
3. Major Issue
4. Catastrophic Issue

By systematically applying these heuristics, we will be able to identify usability issues in the prototype and determine their severity. This process will guide us in improve the design to better meet the needs of our users.

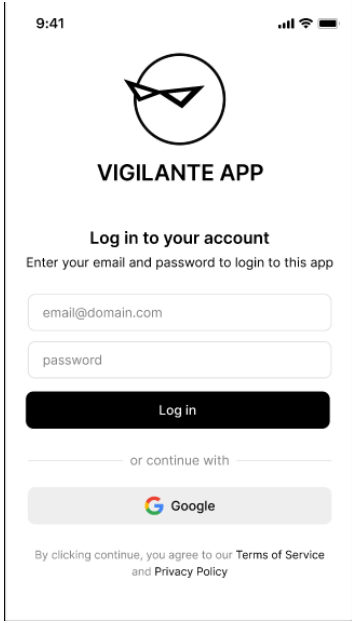
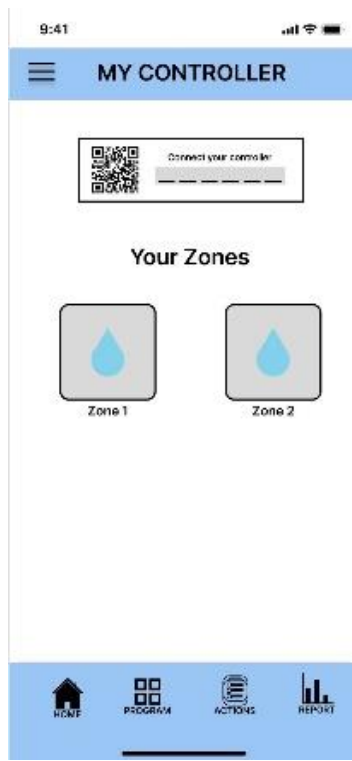
2.0 Method of evaluation

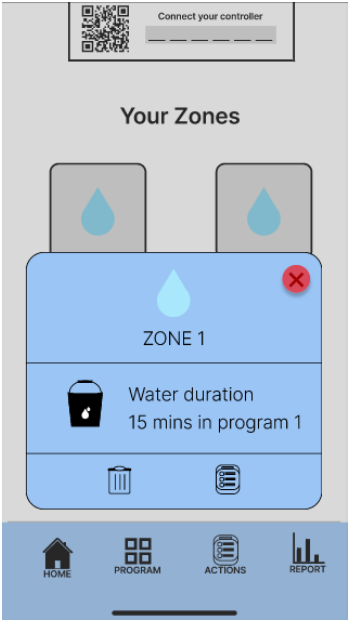
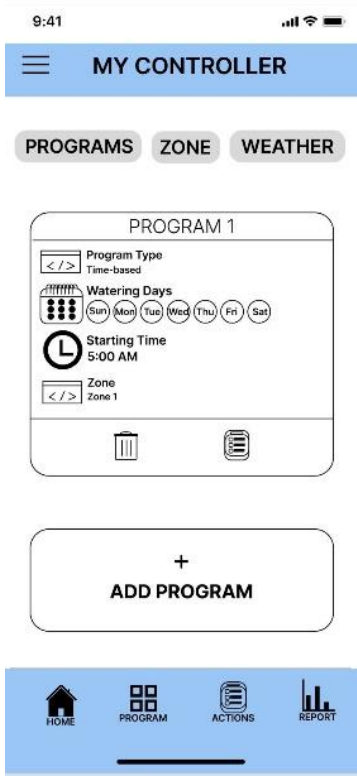
Our group will conduct a heuristic evaluation of Group 12's prototype using a structured approach. First, each member will thoroughly study 10 rules of Nielsen's heuristics and associated severity ratings to ensure a solid understanding of the evaluation criteria. Then, we will independently review the prototype, systematically applying Nielsen's heuristics to identify any usability issues.

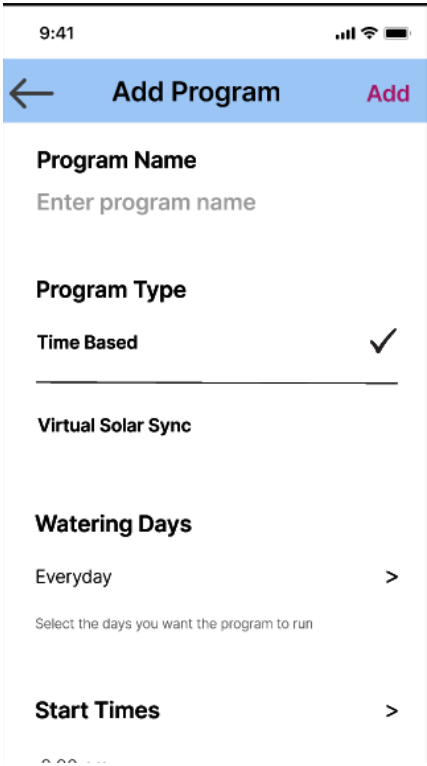
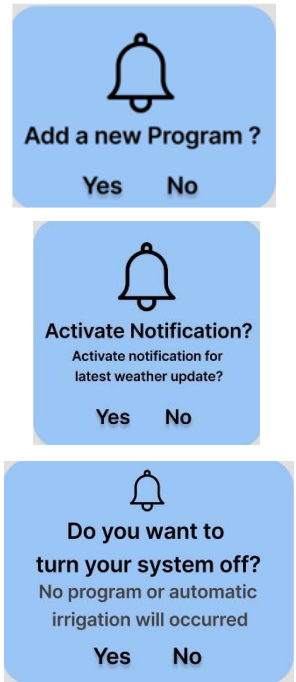
After completing our individual evaluations, we will compile and compare our findings, discussing each identified issue to reach a consensus on their validity and severity. Using our collective feedback, we will brainstorm and generate ideas for improving the design, focusing on addressing the identified issues and enhancing the overall user experience. The revised prototype will then be tested with random users to gather additional insights and validate the effectiveness of the improvements. This feedback will inform further improvements, ensuring the design meets user needs and expectations. The result of the updated design will then be used to test on random users for further findings. By following this methodical process, we aim to create a user-friendly and efficient design for our student feedback system.

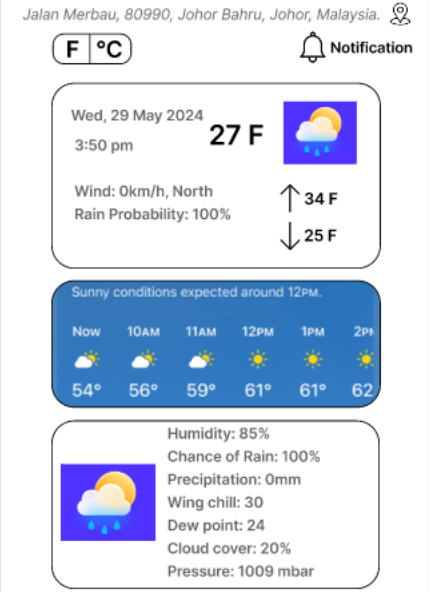
The persona for our evaluation is Lee Yin Shen, who wants to reduce water wastage caused by overwatering his plants after being unaware of the rain when he was away. His goal is to ensure efficient watering of plants, avoiding overwatering, and saving water. The three main user tasks identified for this evaluation are using the smart watering system, checking weather reports, and managing data.

3.0 Heuristic Evaluation (HE) Table

No	Prototype Image	Identified Issue	Heuristic and Severity
1		If I forgot the password or want to reset the password, there are no option for me to reset the password.	H9: Help users recognize, diagnose, recover from errors. S4: Catastrophic issue
2		There is no option to add new zone at the homepage, the user need to manage the zone at the controller part, which is not convenient.	H3: User control and freedom S2: Minor issue

3		After select the zone, the delete button is not function, which means I can't delete the zone.	H3: User control and freedom S3: Major issue
4		For the existing program, there are no option to edit the program, so that the user need to add a new one which is not convenient.	H7: Flexibility and efficiency of use S3: Major issue

5		For program, there is no option for the user to change the watering mode, this may cause limitation to the watering process.	H7: Flexibility and efficiency of use S3: Major issue
6		The bell icon is repeatedly used for many process, including add new program, activate notification, turn off system, and notification for successfully turn off the system. User might get confused of the function without reading the text.	H4: Consistency and standards. S3: Major issue

7		The layout lacks a clear visual hierarchy, making it difficult for users to quickly find the most relevant information. For example, the chance of rain text is too small.	H8: Aesthetic and minimalist design S2: Minor issue
8		For report, there should be an option to generate report based on date. Another option of yearly report also can be added.	H7: Flexibility and efficiency of use S3: Major issue

9		The report doesn't show any graph or data, so this report function is not working.	H4: Visibility of system status S4: Catastrophic issue
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4.0 User journey map



User: Lee Yin Shen

Scenario: Mr. Lee Yin Shen wants to reduce water wastage caused by overwatering his plants after being unaware of the rain when he was away.

User goal: Ensure efficient watering of plants, avoiding overwatering and saving water.

TASK 1: Smart Watering System	TASK 2: Checking Weather Reports	TASK 3: Managing Data
1.Open Vigilante Farm app 2.Sign in 3.Click Zone 1 4.Click Actions icon 5.Click Start Program 6.Click Program 1 7.Click Start 8.Monitor the program in home page	9. Click Program icon 10.Click Weather 11.Set the format of temperature to °C	12.Click Report icon 13.Click Daily Report 14.Close application
“Wow! I love how I can control the time to water my plants and the area I want to water while being away from home”	“Oh no! It is starting to rain while I was away from home but I watered my plants this morning. I didn’t notice the chance of rain later is 100%.”	“Wow! I can save the data in this app. I do not have to go to Excel to save my data again, which is very convenient.”
Painpoints: 1.No reset the password option on log in page 2.No option to add new zone at the homepage 3.The delete button after clicking Zone 1 does not function	Painpoints: 1.The chance of rain text in the weather page is too small to be noticed immediately Opportunities: 1.Improve UI of weather page to allow the user to notice the chance of rain text immediately	Painpoints: 1.No option to generate report based on date 2.The report function is not working as it doesn’t show any graph or data Opportunities:

4.No option to edit the program added previously 5.After clicking add program, there is no option for the user to change the watering days mode Opportunities: 1.Add reset the password option on log in page 2.Put another add new zone icon at the homepage 3.Ensure the delete button after clicking Zone 1 function 4.Add option to edit the program added previously 5.After clicking add program, enable user to change the watering days mode		1.Add option to generate report based on date 2.Automatically show graph of the user's data after clicking daily report, weekly report or monthly report.
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